AI ASSISTANT CODING ASSIGNMENT-14.5

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Batch:04

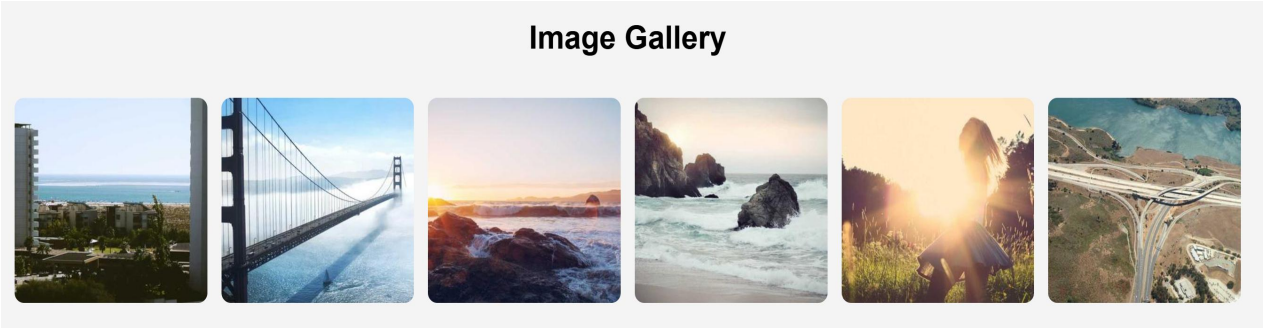
Task 1 – Image Gallery Website
Create an image gallery web page.
Requirements:
• Display images in a grid layout.
• Use AI to:
• Optimize layout using CSS Grid.
• Add hover zoom or overlay effects.
• Ensure mobile responsiveness.

Prompt: Create a responsive image gallery using HTML and CSS Grid with a hover zoom effect.

Code:

```html
<!DOCTYPE html>
<html>
<head>
<title>Image Gallery</title>
<style>
body{
font-family: Arial;
text-align:center;
background: #f4f4f4;
}
.gallery{
display:grid;
grid-template-columns:repeat(auto-fit,minmax(200px,1fr));
gap:15px;
padding:20px;
}
.gallery img{
width:100%;
border-radius:10px;
transition:0.3s;
}
.gallery img:hover{
transform:scale(1.1);
}
</style>
</head>
<body>
<h1>Image Gallery</h1>
<div class="gallery">
<img src="https://picsum.photos/300?1">
<img src="https://picsum.photos/300?2">
<img src="https://picsum.photos/300?3">
<img src="https://picsum.photos/300?4">
<img src="https://picsum.photos/300?5">
<img src="https://picsum.photos/300?6">
</div>
</body>
</html>
```

Output:



Explanation:
The image gallery displays images using **CSS Grid layout**.
The grid automatically adjusts columns depending on screen size, making it responsive.
A **hover zoom effect** is applied using CSS transform to enlarge images when the mouse pointer moves over them.

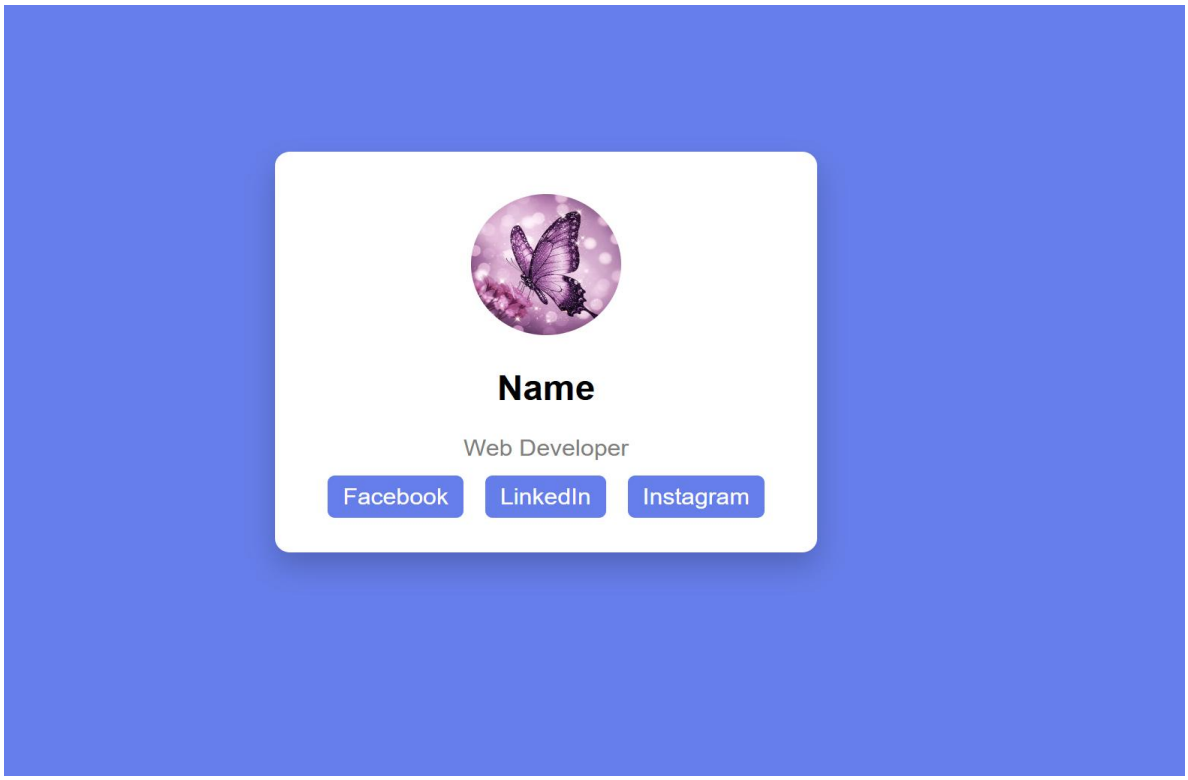
Task 2 – Profile Card Generator
Create a profile card web page.
Requirements:
• Display user image, name, role, and social media links.
• Use AI to:
• Suggest modern UI colors and fonts.
• Design the card using Flexbox.
• Add hover animation and shadow effects

Prompt: Create a modern profile card using HTML and CSS Flexbox with hover animation.

Code:

```html
<!DOCTYPE html>
<html>
<head>
<title>Profile Card</title>
<style>
body{
font-family: Arial;
background: #667eea;
display:flex;
justify-content:center;
align-items:center;
height:100vh;
}
.card{
background: white;
padding:30px;
text-align:center;
border-radius:10px;
box-shadow:0 10px 20px rgba(0,0,0,0.2);
transition:0.3s;
}
.card:hover{
transform:translateY(-10px);
}
.profile{
width:100px;
border-radius:50%;
}
.role{
color: gray;
}
.social a{
text-decoration:none;
margin:5px;
padding:6px 10px;
background: #667eea;
color: white;
border-radius:5px;
}
</style>
</head>
```

Output:



Explanation:
The profile card shows a **user image, name, role, and social media links**.
The layout is styled with modern colors and uses **Flexbox to center the card**.
A **hover animation** moves the card slightly upward and a **shadow effect** gives it a modern UI appearance.

Task 3 -Develop a dynamic shopping cart system.
Requirements:
• Add, remove, and update product quantities.
• Display real-time total price.
• Use AI to:
•Manage cart state using JavaScript objects.
• Persist cart data in localStorage.

Prompt: Build a dynamic shopping cart using HTML, CSS, and JavaScript with localStorage.

Code:

```
<!DOCTYPE html>
<html>
<head>
<title>Shopping Cart</title>
<style>
body{
font-family:Arial;
text-align:center;
background: #f4f4f4;
}
.product{
background: white;
padding:15px;
margin:10px;
display:inline-block;
border-radius:10px;
box-shadow:0 5px 10px rgba(0,0,0,0.2);
}
button{
padding:5px 10px;
margin:5px;
background: #4CAF50;
color: white;
border:none;
border-radius:5px;
cursor:pointer;
}
</style>
</head>
<body>
<h1>Shopping Cart</h1>
<div class="product">
<h3>Laptop</h3>
<p>$800</p>
<button onclick="addItem('Laptop',800)">Add</button>
</div>
<div class="product">
<h3>Phone</h3>
<p>$500</p>
<button onclick="addItem('Phone',500)">Add</button>
</div>
```

Output:

Explanation:
The shopping cart allows users to **add products and view them in the cart**.
JavaScript objects store product details such as name, price, and quantity.
The **total price updates dynamically** whenever products are added or removed.
Cart data is stored in **localStorage**, ensuring that items remain saved even after refreshing the page.

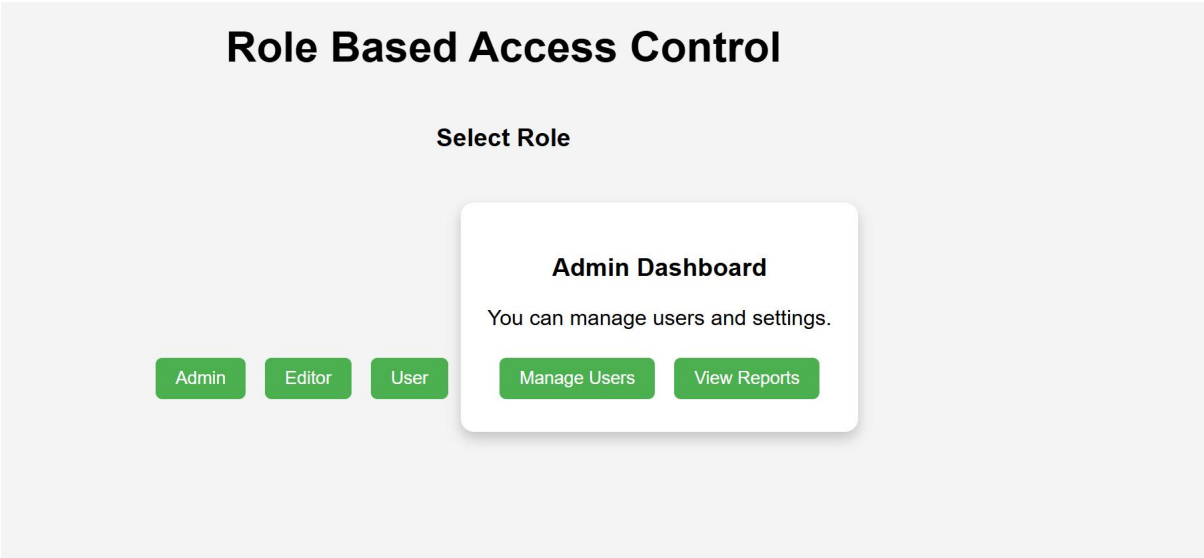
Task 4 – Role-Based Access Control Application
Create a role-based web application.
Requirements:
• Admin, Editor, User roles.
• Conditional UI rendering.
• Use AI to:
• Design access control logic.
• Maintain secure state transitions.
• Create scalable UI components.

Prompt: Create a role-based web app with Admin, Editor, and User roles using JavaScript.

Code:

```html
<!DOCTYPE html>
<html>
<head>
<title>Role Based Access Control</title>
<style>
body{
font-family: Arial;
text-align:center;
background: #f4f4f4;
}
.container{
margin-top:40px;
}
button{
padding:8px 15px;
margin:5px;
background: #4CAF50;
color: white;
border:none;
border-radius:5px;
cursor:pointer;
}
.panel{
margin-top:20px;
padding:20px;
background: white;
display:inline-block;
border-radius:10px;
box-shadow:0 5px 10px rgba(0,0,0,0.2);
}
</style>
</head>
<body>
<h1>Role Based Access Control</h1>
<div class="container">
<h3>Select Role</h3>
<button onclick="setRole('Admin')">Admin</button>
<button onclick="setRole('Editor')">Editor</button>
<button onclick="setRole('User')">User</button>
<div class="panel" id="panel">
<p>Please select a role</p>
```

Output:



Explanation:

This application demonstrates **role-based access control with Admin, Editor, and User roles**.
Each role sees **different interface options** using conditional UI rendering.
JavaScript controls the access logic to ensure users only see features allowed for their role.

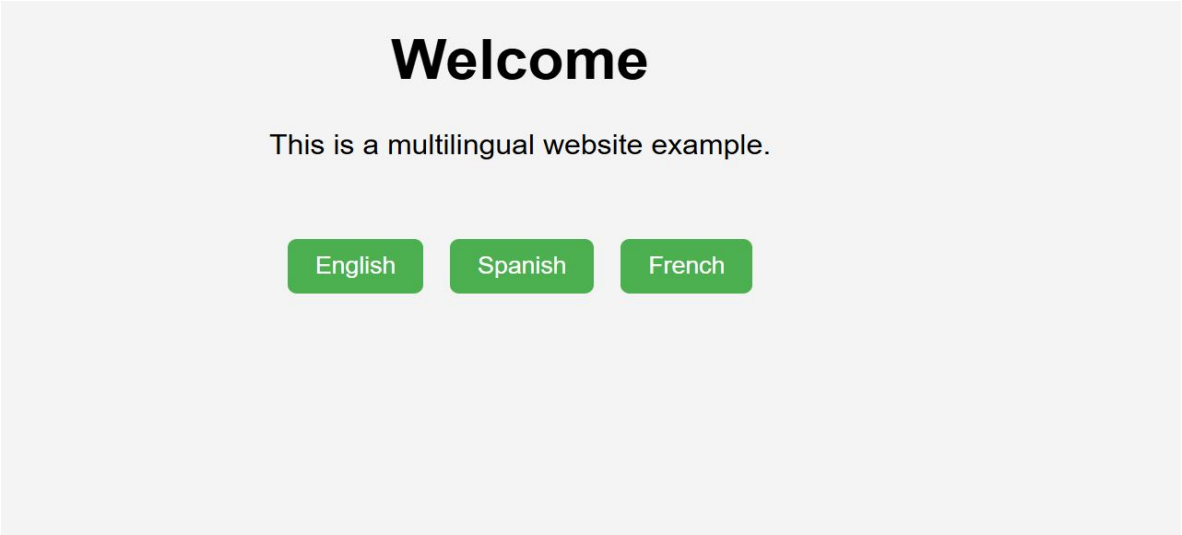
Task 5 – Multi-Language Website
Design a multilingual website.
Requirements:
• Switch languages dynamically.
• Persist language preference.
• Use AI to:
•Structure translation files.
•Handle RTL/LTR layouts.
•Improve accessibility

Prompt: Create a multilingual website with dynamic language switching using JavaScript.

Code:

```
<!DOCTYPE html>
<html>
<head>
<title>Multi Language Website</title>
<style>
body{
font-family: Arial;
text-align:center;
background: #f4f4f4;
}
.container{
margin-top:40px;
}
button{
padding:8px 15px;
margin:5px;
background: #4CAF50;
color: white;
border:none;
border-radius:5px;
cursor:pointer;
}
</style>
</head>
<body>
<h1 id="title">Welcome</h1>
<p id="text">This is a multilingual website example.</p>
<div class="container">
<button onclick="setLanguage('en')">English</button>
<button onclick="setLanguage('es')">Spanish</button>
<button onclick="setLanguage('fr')">French</button>
</div>
<script>
const translations = {
en:{
title:"Welcome",
text:"This is a multilingual website example."
},
es:{
title:"Bienvenido",
text:"Este es un ejemplo de sitio web multilingüe."
```

Output:



Explanation:
This multilingual website allows users to **switch between different languages dynamically**.
Translations are stored in **JavaScript objects**, making it easy to manage multiple languages.
The selected language is saved using **localStorage**, so the preference remains after refreshing the page.
This approach improves **accessibility and usability** for users from different language backgrounds.

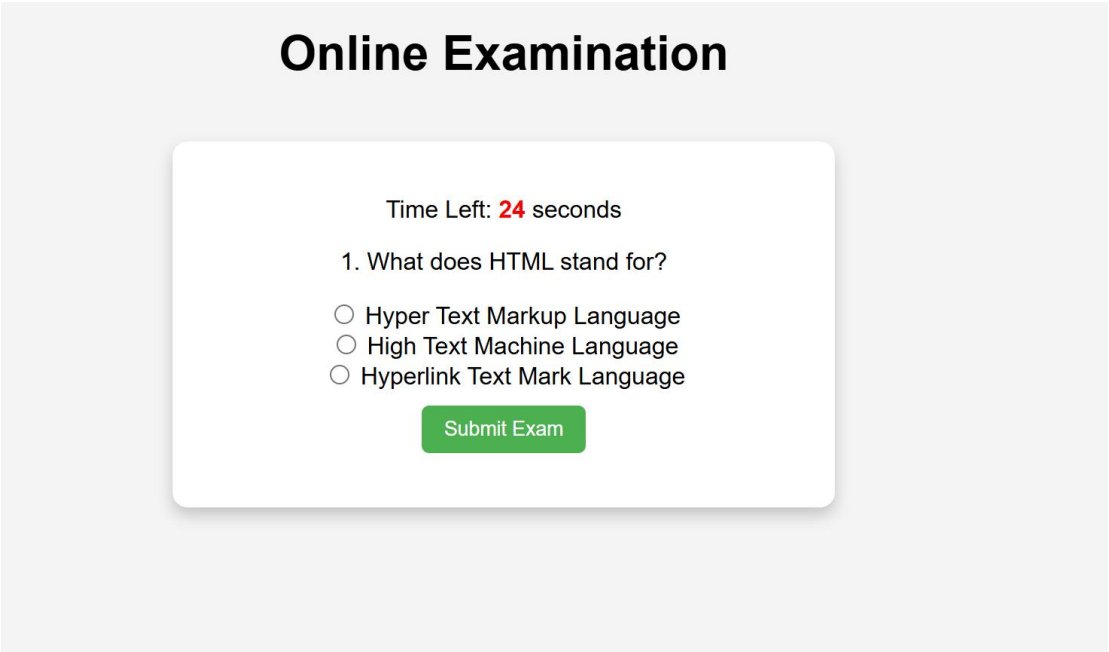
Task 6 – Online Examination System
Create an online exam interface.
Requirements:
• Timer, questions, and submission.
• Prevent multiple submissions.
• Use AI to:
•Manage exam state.
•Improve exam UX.
•Ensure accessibility.

Prompt: Create an online exam interface with questions, timer, and submission using HTML, CSS, and JavaScript.

Code:

```html
<!DOCTYPE html>
<html>
<head>
<title>Online Exam</title>
<style>
body{
font-family: Arial;
text-align:center;
background: #f4f4f4;
}
.exam-box{
background: white;
padding:20px;
margin:40px auto;
width:400px;
border-radius:10px;
box-shadow:0 5px 10px rgba(0,0,0,0.2);
}
button{
padding:8px 15px;
margin-top:10px;
background: #4CAF50;
color: white;
border:none;
border-radius:5px;
cursor:pointer;
}
#timer{
font-weight:bold;
color: red;
}
</style>
</head>
<body>
<h1>Online Examination</h1>
<div class="exam-box">
<p>Time Left: <span id="timer">30</span> seconds</p>
<p>1. What does HTML stand for?</p>
<label><input type="radio" name="q1"> Hyper Text Markup Language</label><br>
<label><input type="radio" name="q1"> High Text Machine Language</label><br>
<label><input type="radio" name="q1"> Hyperlink Text Mark Language</label><br>
```

Output:



Explanation:

This online examination system displays questions with a **countdown timer**. Users can answer the question and submit the exam. The timer automatically submits the exam when time ends. JavaScript manages the **exam state and prevents multiple submissions**, improving the user experience and accessibility.